Manifold LLC

LOUDOUN COUNTY, VIRGINIA · AI/ML ADVICE AND ENGINEERING

Independent AI/ML advice and engineering for intelligence and defense programs.

A small Virginia firm of cleared practitioners with about ten years each inside Intelligence Community acquisition and research programs. We help program offices decide whether to build or buy, and when the answer is build, we build. Never both for the same organization.

TS/SCI WITH CI POLYGRAPH SMALL BUSINESS · UEI C6P2G6PQLEB8 · CAGE 23X60 SENIOR PRACTITIONERS DO THE WORK

WHAT WE DO

Strategy and roadmaps 01

Where foundation models fit in a portfolio, what they cost to run, and what to buy rather than build.

Acquisition and compute 02

Requirements vendors can bid against. GPU and TPU cluster sizing and cost models for government cloud or your own hardware.

Program assessment 03

An outside technical read on an AI program: where it stands, what finishing takes, whether the metrics mean what they seem.

Secure generative AI 04

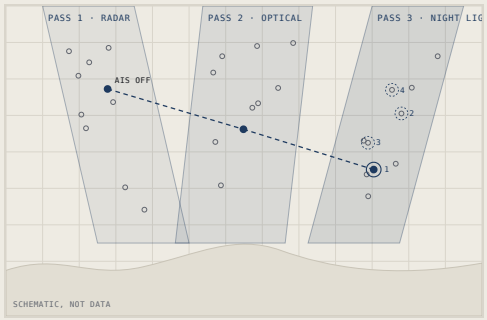
Language-model systems for CUI at IL4 and IL5: reference architectures, retrieval-augmented generation, evaluation and bias audits.

Applied research and development 05

Detectors and trackers over satellite imagery, foundation models on sponsor data, and the GPU and TPU kernels that make them affordable.

How we work

The practitioner who scopes the work does it and briefs it. Advise or build, never both for one organization; no vendor fees. Classified work on customer premises.



Linking a ship that switched off AIS across three satellite passes. Schematic, not data.

SELECTED WORK OF THE FOUNDERS

Dark vessels — A multi-sensor tracker that re-finds ships that switched off AIS from one satellite pass to the next, scored against AIS it never saw.

FastSAR — Open-source radar image formation on CPUs, GPUs and TPUs, faster than every open-source code tested at equal or better accuracy. GitHub, PyPI, arXiv.

Secure generative AI — A production retrieval-augmented generation system for CUI on Amazon Bedrock in IL4 AWS GovCloud, at a Navy UARC.

Geospatial foundation models — Architects for a sponsor's foundation-model effort over overhead data, alongside Google and NVIDIA engineering teams.

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Capability statement with past performance and company data: manifold-llc.com/manifold-capability-statement.pdf

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